

Assignment1-Preemption 报告

Checkpoint 1

结果截图

```
[WARN 0,2] worker: thread 2: exiting
[sched 0,2] sched: switch to scheduler pid(2)
[sched 0,-1] scheduler: switch to proc pid(3)
[ERROR 0,3] kernel_trap: kerneltrap: not from supervisor mode
[ERROR 0,3] kernel_trap: ===== Kernel Panic =====
sstatus : 0x80000000200006020
- SUM:0, SPP:U, SPIE:1, SIE: 0
scause : 0x0000000000000002
- Interrupt:0, Code:2
sepc : 0x000000008020197c
stval : 0x00000000100024f3
sip : 0x0000000000000000
- Pending: Software:0, Timer:0, External:0
sie : 0x0000000000000220
- Enabled: Software:0, Timer:1, External:1
satp : 0x0000000000000000
kernel trapframe at 0x000000008023ce60
ra: 0x0000000080202364 sp: 0x000000008023ce60 gp: 0x0000000000000000 tp: 0x0000000000000000
t0: 0x0000000000000000 t1: 0x000000008023cf48 t2: 0x0000000000000100 s0: 0x000000008023cf80
s1: 0x0000000000000000 a0: 0x00000000000001f4 a1: 0x000000000000000a a2: 0x0000000000000005
a3: 0x00000000004c4b40 a4: 0x00000000b17bc37 a5: 0x0000000080202364 a6: 0x00000000802062c0
a7: 0x0000000000000000 s2: 0x0000000000000000 s3: 0x00000000000003e8 s4: 0x0000000000000003
s5: 0x0000000080205ad8 s6: 0x00000000802054b0 s7: 0x0000000080206300 s8: 0x0000000000000000
s9: 0x0000000080238d98 s10: 0x0000000000000000 s11: 0x0000000000000000 t3: 0x0000000000000010
t4: 0x000000008046dda t5: 0x000000000000000f t6: 0x0000000000000027

[PANIC 0,3] os/trap.c:93: kernel panic
```

说明

- 本题完成了基础的抢占/调度验证流程。
- 从运行结果看，系统能够正常进入并处理时钟中断，调度路径可达，程序行为符合 checkpoint1 预期。

Checkpoint 2

结果截图

```
oslab@oslab-2025S:~/Desktop/Assignment/assignment1-Preemption$ set -e; OUT1="/home/oslab/.vscode-server/data/User/workspaceStorage/c9f1095a75f455f68a5919443baa
4c81/GitHub.copilot-chat/chat-session-resources/37e6a7f2-1e5d-4b22-97ee-917ed993af62/call_t5FM31401YJAGphKuojf1cFu__vscode-1774850053574/content.txt"; grep -nE
"init: init ends|init process exited|kerneltrap: not from supervisor mode|Kernel Panic" "$OUT1" || true
345:[INFO 0,1] init: init ends!
346:[PANIC 0,1] os/proc.c:225: init process exited
```

说明

- 问题根因是：在内核 Trap 中触发调度时，没有成对保护/恢复 `sstatus` 与 `sepc`，导致后续 `sret` 使用了被其他上下文影响后的 CSR 状态。
- 由于 `sret` 会依赖 `SPP` 位决定返回级别，若该位状态不正确，就可能出现异常返回（例如错误回到 U 模式），从而触发 `SPP:U` 类问题。
- 因此 checkpoint2 的关键是保证 Trap 内调度场景下的上下文一致性，避免 CSR 状态被“串台”。
- 为什么“总在第一个进程 exit 后更容易出现”：exit 会触发一次明确的 sched 切换；同时其它进程很多都停在 kernel_trap 的 yield 处，这会制造大量“Trap 内调度 -> 回来继续 sret”的场景；因此更容易踩到被污染/过期的 `sstatus.SPP`。

Checkpoint 3

结果截图

```
[INFO 3,1] init: all threads exited, count 150000
[INFO 3,1] init: init ends!
[PANIC 3,1] os/proc.c:225: init process exited
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 3,1] init: all threads exited, count 149662
[INFO 3,1] init: init ends!
[PANIC 3,1] os/proc.c:225: init process exited
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 1,1] init: all threads exited, count 150000
[INFO 1,1] init: init ends!
[PANIC 1,1] os/proc.c:225: init process exited
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 2,1] init: all threads exited, count 150000
[INFO 2,1] init: init ends!
[PANIC 2,1] os/proc.c:225: init process exited
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 0,1] init: all threads exited, count 150000
[INFO 0,1] init: init ends!
[PANIC 0,1] os/proc.c:225: init process exited
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 0,1] init: all threads exited, count 150000
[INFO 0,1] init: init ends!
[PANIC 0,1] os/proc.c:225: init process exited
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 2,1] init: all threads exited, count 149563
[INFO 2,1] init: init ends!
[PANIC 2,1] os/proc.c:225: init process exited
[PANIC 1,-1] os/trap.c:45: other CPU has panicked
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 1,1] init: all threads exited, count 150000  
[INFO 1,1] init: init ends!  
[PANIC 1,1] os/proc.c:225: init process exited  
[PANIC 0,-1] os/trap.c:45: other CPU has panicked  
[PANIC 3,-1] os/trap.c:45: other CPU has panicked  
[PANIC 2,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 3,1] init: all threads exited, count 150000  
[INFO 3,1] init: init ends!  
[PANIC 3,1] os/proc.c:225: init process exited  
[PANIC 2,-1] os/trap.c:45: other CPU has panicked  
[PANIC 1,-1] os/trap.c:45: other CPU has panicked  
[PANIC 0,-1] os/trap.c:45: other CPU has panicked
```

```
[INFO 1,1] init: all threads exited, count 150000  
[INFO 1,1] init: init ends!  
[PANIC 1,1] os/proc.c:225: init process exited  
[PANIC 0,-1] os/trap.c:45: other CPU has panicked  
[PANIC 2,-1] os/trap.c:45: other CPU has panicked  
[PANIC 3,-1] os/trap.c:45: other CPU has panicked
```

说明

- checkpoint3 主要体现了并发场景下计数结果的统计特征。
- `count` 大多数情况下是 15000，因为在理想执行中每次自增都被正确保留，最终总和稳定。
- 少数情况下不是 15000，是因为并发下存在竞争（读-改-写交错、抢占导致的丢失更新），因此会出现小概率偏差。

完成作业耗时

- 约 5 小时。

遇到的困难

- 主要困难在 checkpoint2，需要结合 lab 文档理解 trap 与调度切换细节后才能准确作答。